M-P Neuron

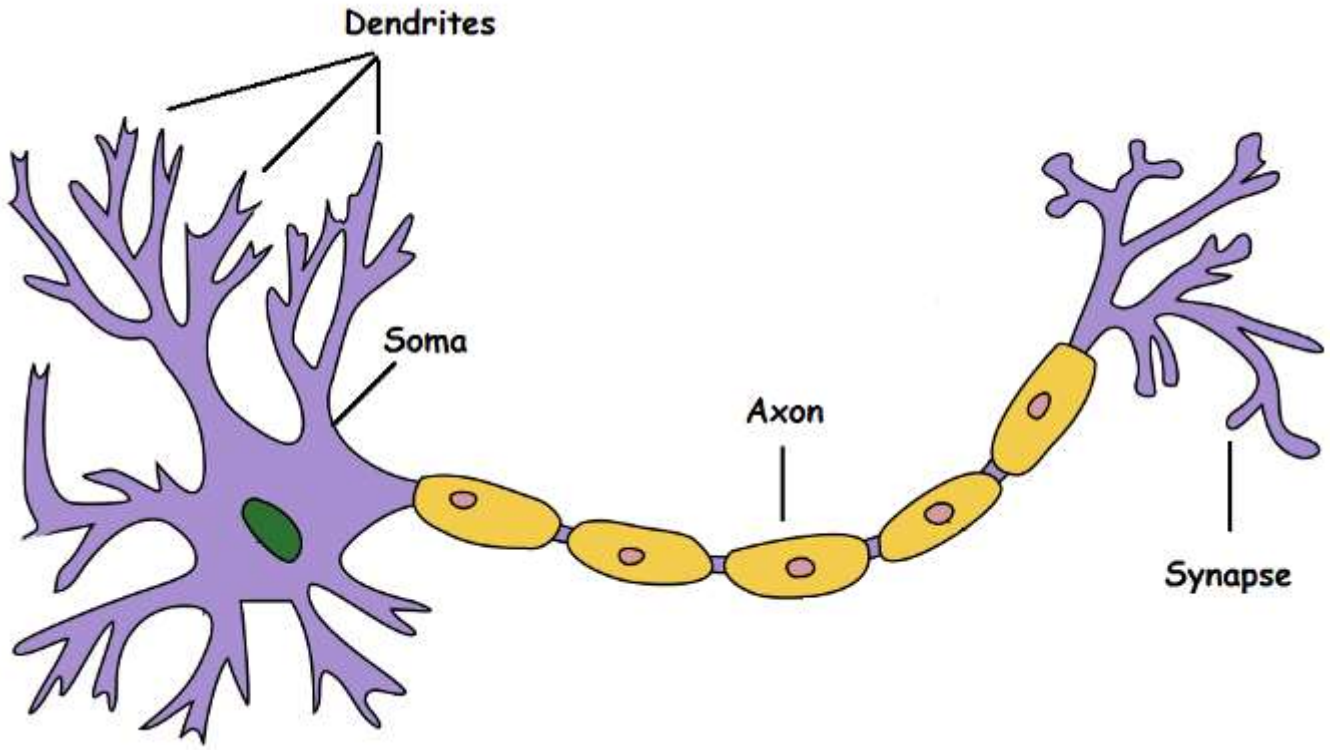
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1. Biological Neuron

Signal Reception:

Dendrites receive neurotransmitters (chemical signals) that come from other neurons. Neurotransmitters create electrical impulses.

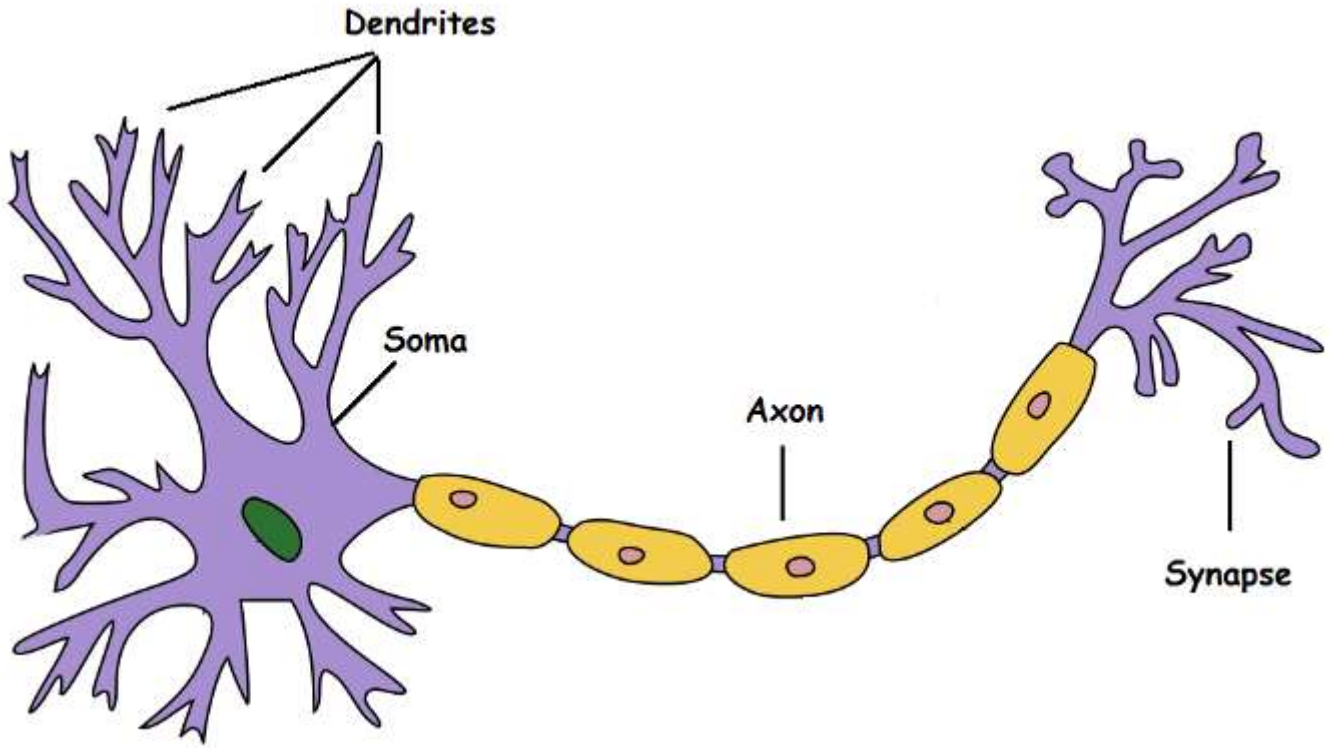


1. Biological Neuron

Signal Integration:

The soma (cell body) processes the incoming signals from the dendrites.

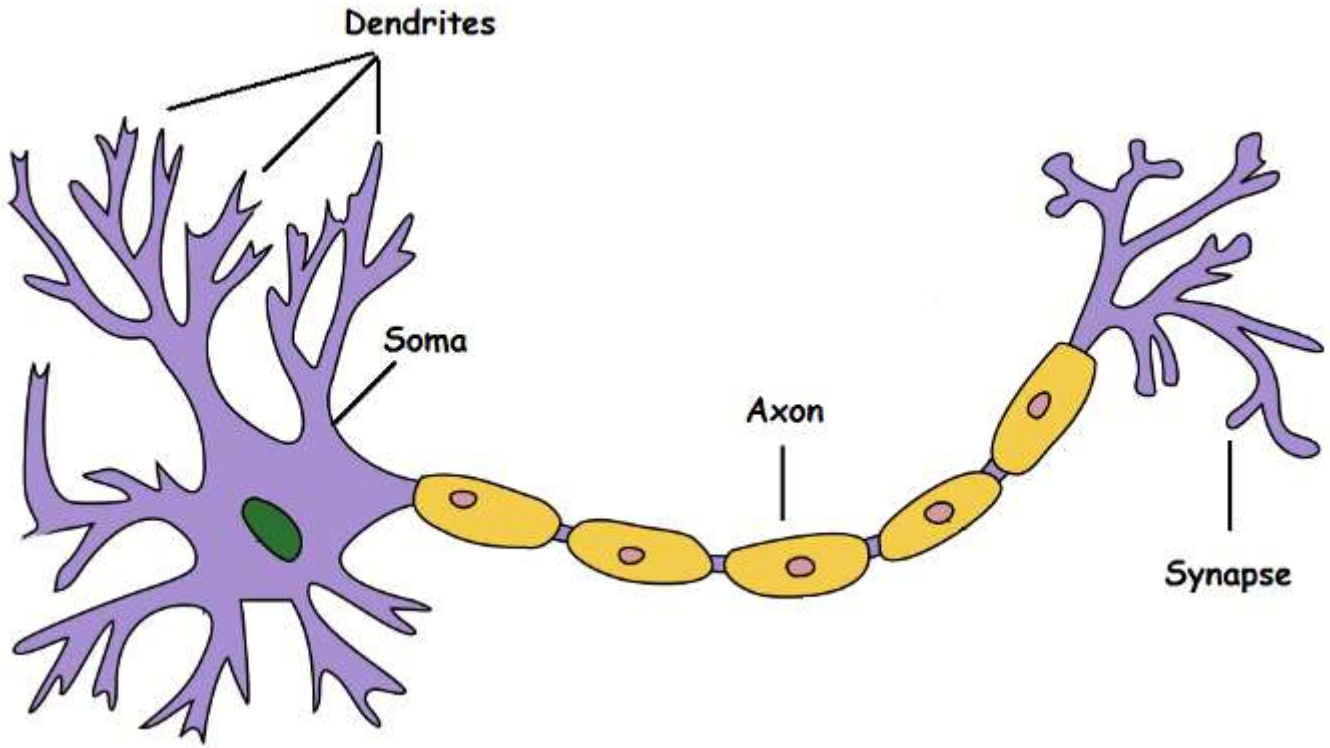
If the combined signals from the dendrites reach a certain threshold, the neuron generates an electrical impulse.



1. Biological Neuron

Signal Transmission:

Once an electrical impulse is generated, it travels down the axon (a long fiber-like structure).

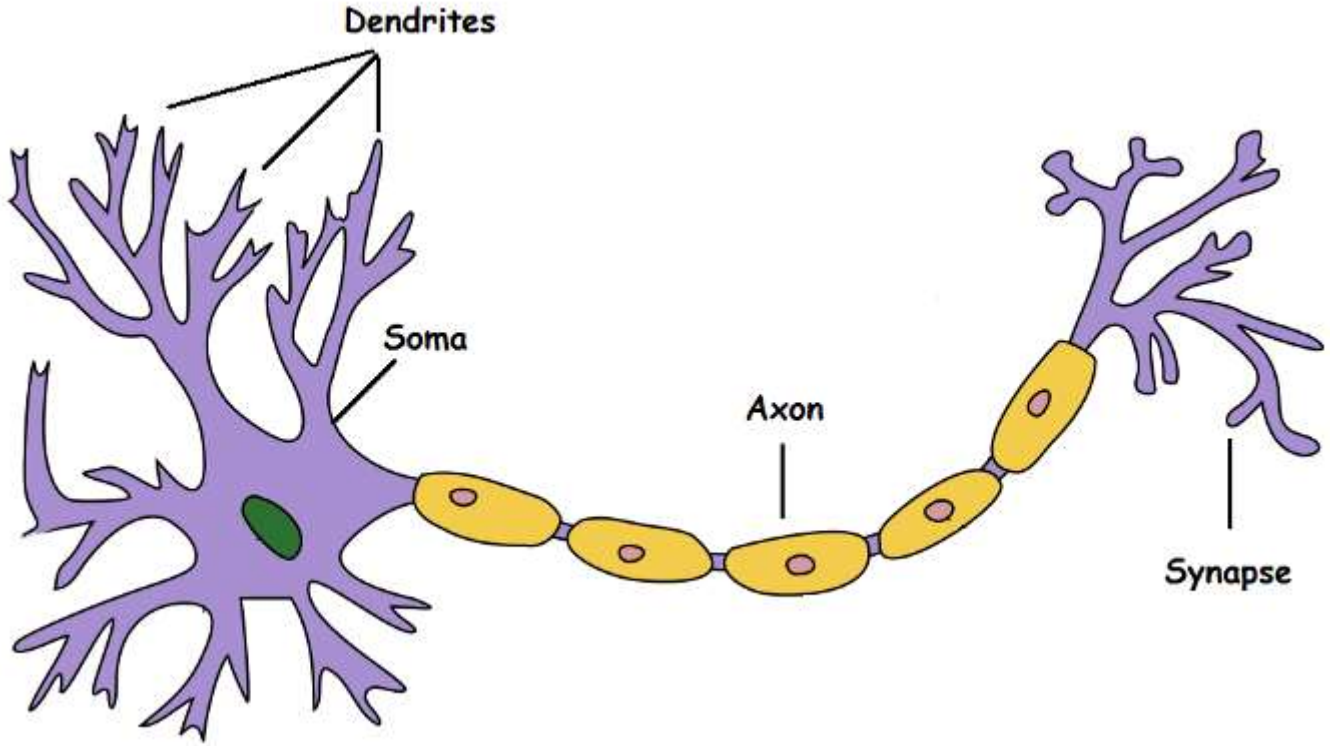


1. Biological Neuron

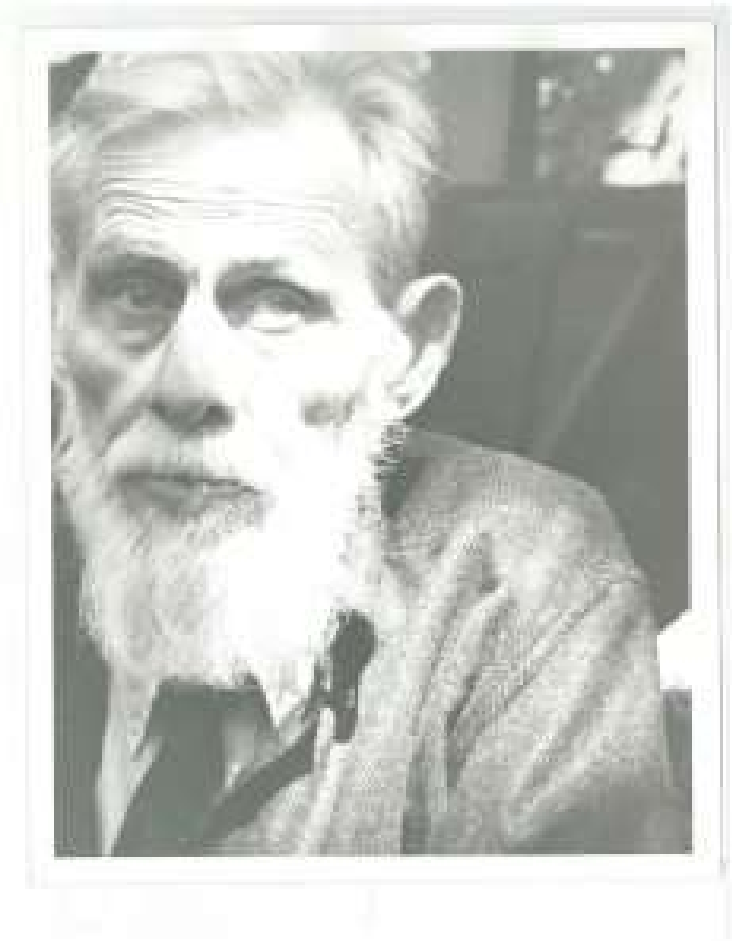
Signal Transfer:

The electrical impulse reaches the synapse, the junction between two neurons.

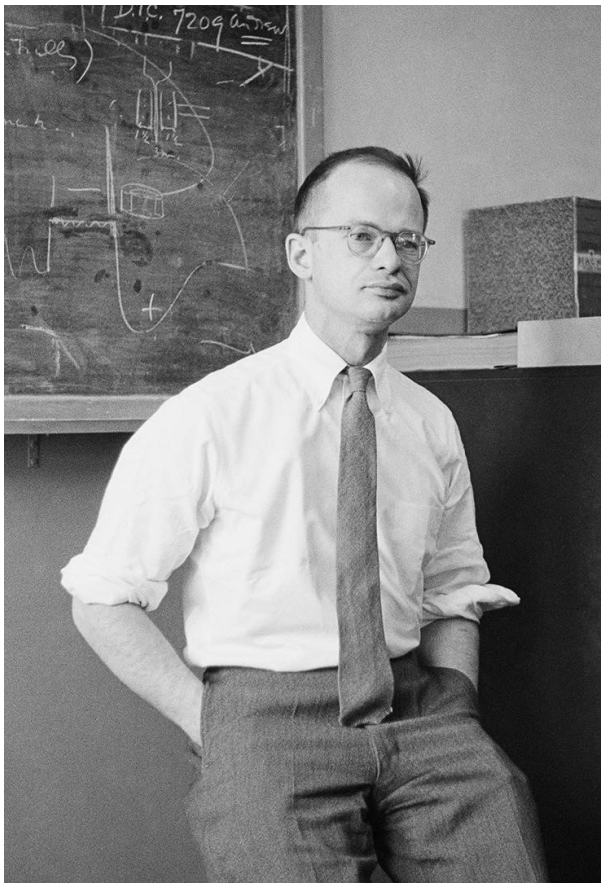
At the synaptic terminal, the electrical signal triggers the release of neurotransmitters (chemical messengers) into the synaptic cleft (gap between neurons).



2. McCulloch-Pitts Neuron (1943)

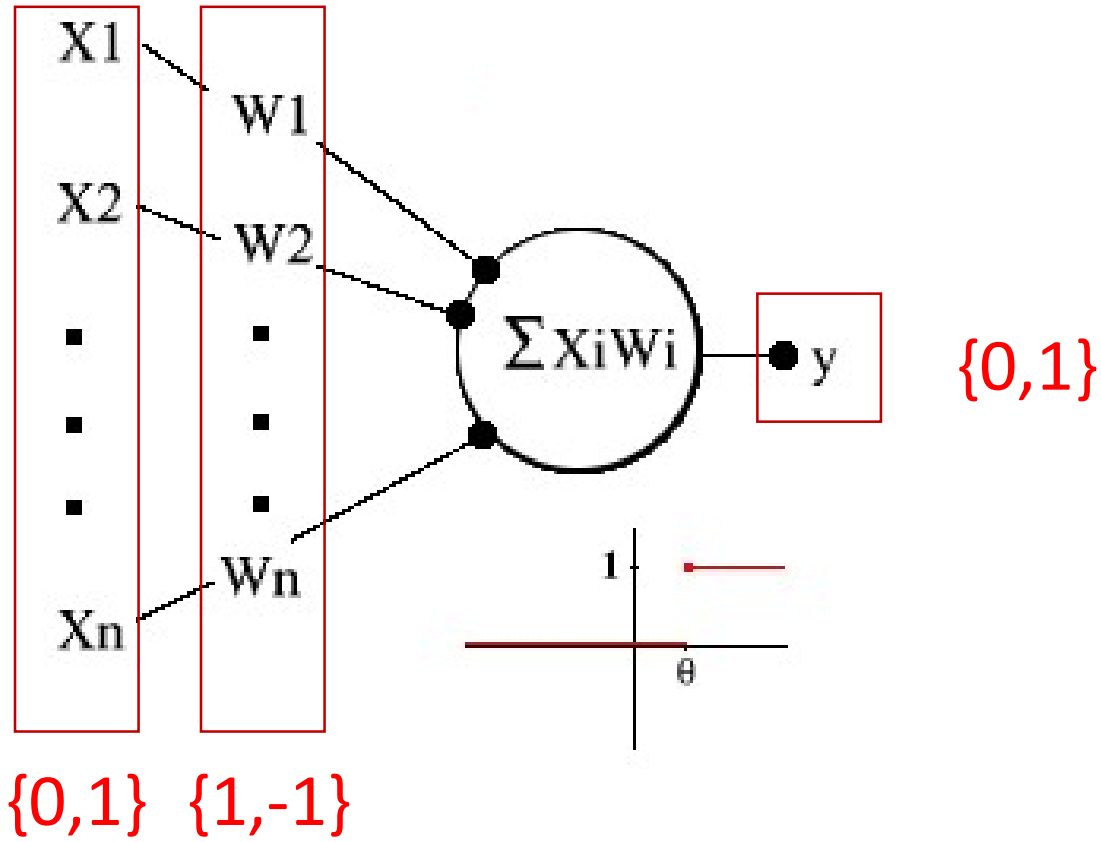


Warren Sturgis McCulloch (1898-1969)



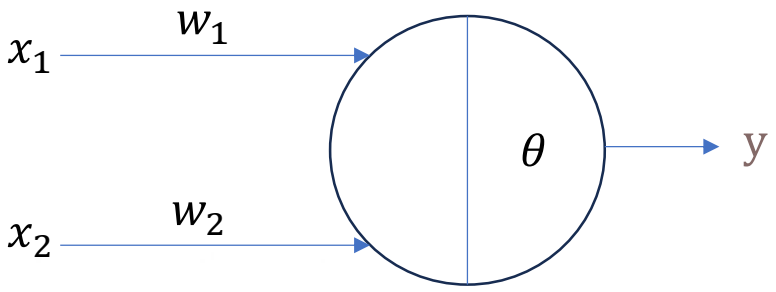
WALTER PITTS (1923-1969)

2. McCulloch-Pitts Neuron



2. McCulloch-Pitts Neuron

M-P Neuron: A Concise Representation

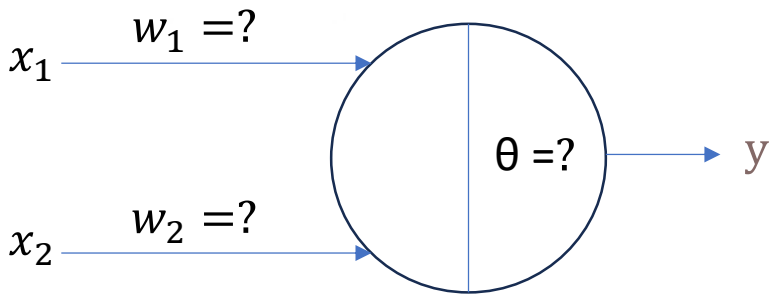


This figure denotes that if the ***weighted sum of Boolean inputs*** x_1 ***and*** x_2 ***is larger or equal to*** θ, **y=1, otherwise, y=0.**

3. Implement Logical operations using M-P Neuron

3.1. AND

- Logic: The output is 1 if all inputs are 1; otherwise, it is 0.



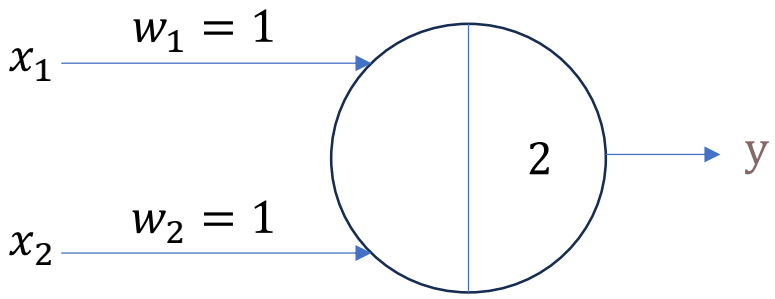
x_1	x_2	∑	y (AND)
0	0		0
0	1		0
1	0		0
1	1		1

3. Implement Logical operations using M-P Neuron

3.1. AND

- Logic: The output is 1 if all inputs are 1; otherwise, it is 0.

- Weights: $w_1 = w_2 = 1$
- Threshold: θ = 2.

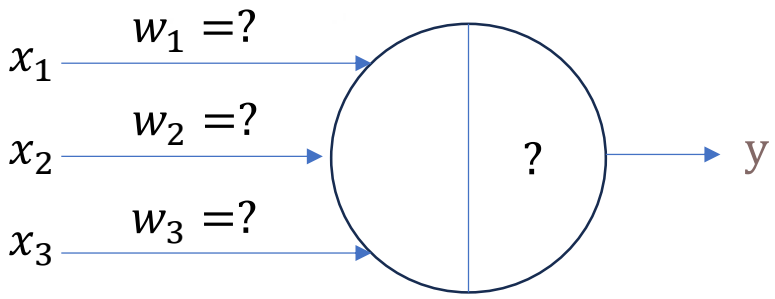


x_1	x_2	∑	y (AND)
0	0	0<2	0
0	1	1<2	0
1	0	1<2	0
1	1	2≥2	1

3. Implement Logical operations using M-P Neuron

3.2. AND (3 inputs)

- Logic: The output is 1 if all inputs are 1; otherwise, it is 0.

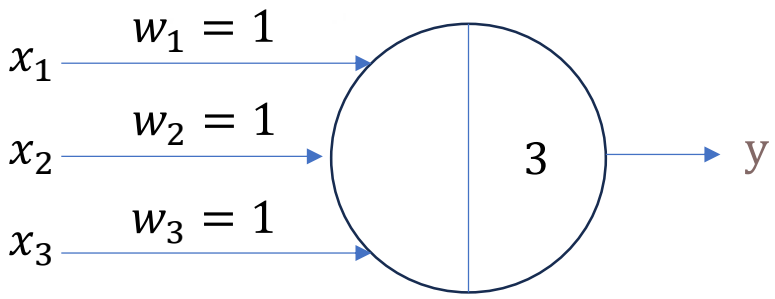


x_1	x_2	x_3	∑	y
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		1

3. Implement Logical operations using M-P Neuron

3.2. AND (3 inputs)

- Logic: The output is 1 if all inputs are 1; otherwise, it is 0.

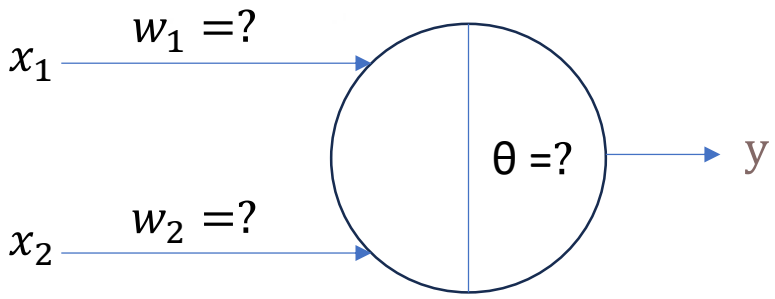


x_1	x_2	x_3	∑	y
0	0	0	0<3	0
0	0	1	1<3	0
0	1	0	1<3	0
0	1	1	2<3	0
1	0	0	1<3	0
1	0	1	2<3	0
1	1	0	2<3	0
1	1	1	3≥3	1

3. Implement Logical operations using M-P Neuron

3.3. OR

- Logic: The output is 1 if at least one input is 1; otherwise, it is 0



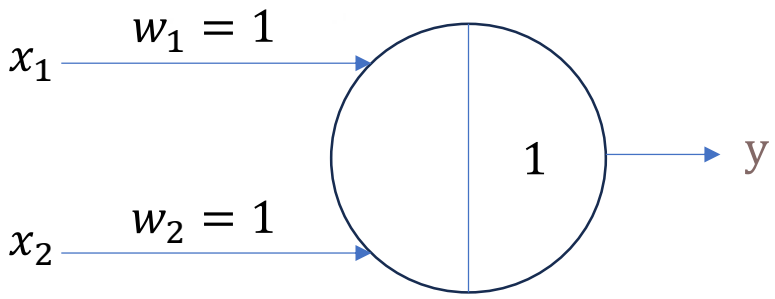
x_1	x_2	∑	y (OR)
0	0		0
0	1		1
1	0		1
1	1		1

3. Implement Logical operations using M-P Neuron

3.3. OR

- Logic: The output is 1 if at least one input is 1; otherwise, it is 0

- Weights: $w_1 = w_2 = 1$
- Threshold: θ = 1.

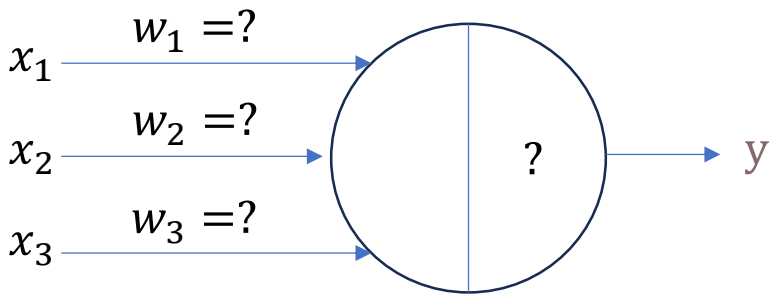


x_1	x_2	∑	y (OR)
0	0	0<1	0
0	1	1≥1	1
1	0	1≥1	1
1	1	2≥1	1

3. Implement Logical operations using M-P Neuron

3.4. OR (3 inputs)

- Logic: The output is 1 if at least one input is 1; otherwise, it is 0.

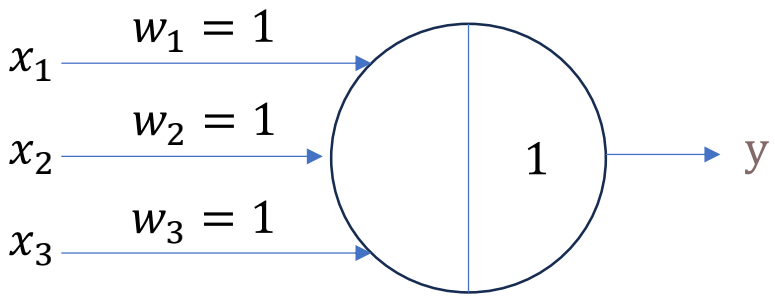


x_1	x_2	x_3	∑	y
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

3. Implement Logical operations using M-P Neuron

3.4. OR (3 inputs)

- Logic: The output is 1 if at least one input is 1; otherwise, it is 0.

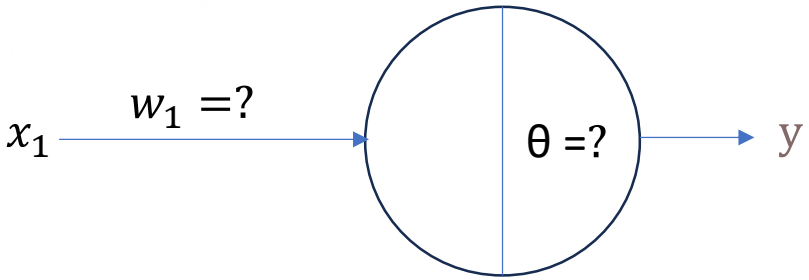


x_1	x_2	x_3	∑	y
0	0	0	0<1	0
0	0	1	1≥1	1
0	1	0	1≥1	1
0	1	1	2≥1	1
1	0	0	1≥1	1
1	0	1	2≥1	1
1	1	0	2≥1	1
1	1	1	3≥1	1

3. Implement Logical operations using M-P Neuron

3.5. NOT

- Logic: The output is 1 if the input is 0; otherwise, it is 0.



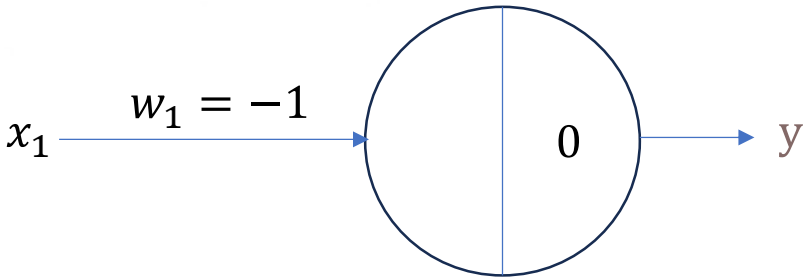
x	∑	y (NOT)
0		1
1		0

3. Implement Logical operations using M-P Neuron

3.5. NOT

- Logic: The output is 1 if the input is 0; otherwise, it is 0.

- Weights: $w_1 = -1$
- Threshold: θ = 0.

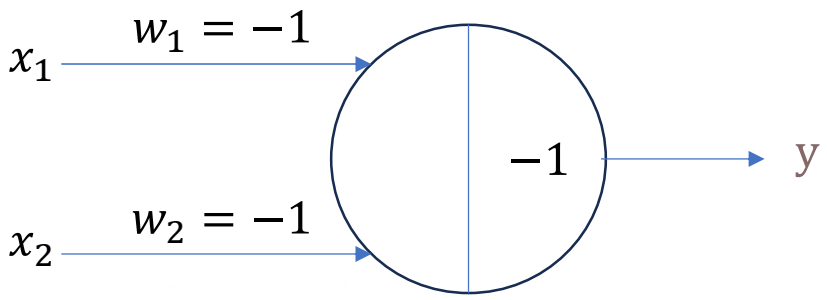


x	∑	y (NOT)
0	0≥0	1
1	−1<0	0

3. Implement Logical operations using M-P Neuron

3.6. NAND（NOT AND）

- Logic: The output is 0 if both inputs are 1; otherwise, it is 1.

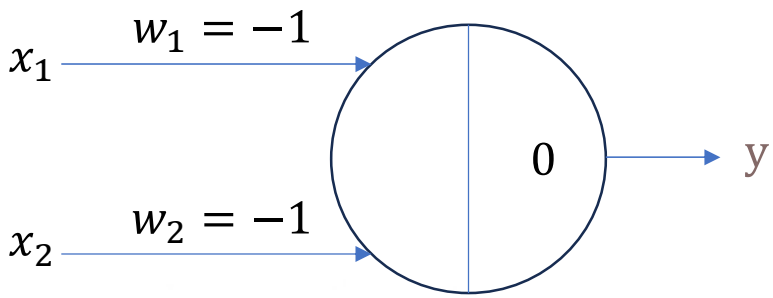


x_1	x_2	y (NAND)
0	0	1
0	1	1
1	0	1
1	1	0

3. Implement Logical operations using M-P Neuron

3.7. NOR（NOT OR）

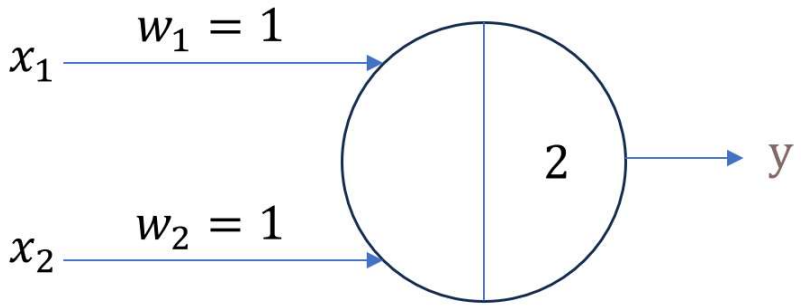
- Logic: The output is 1 if both inputs are 0; otherwise, it is 0.



x_1	x_2	y (NOR)
0	0	1
0	1	0
1	0	0
1	1	0

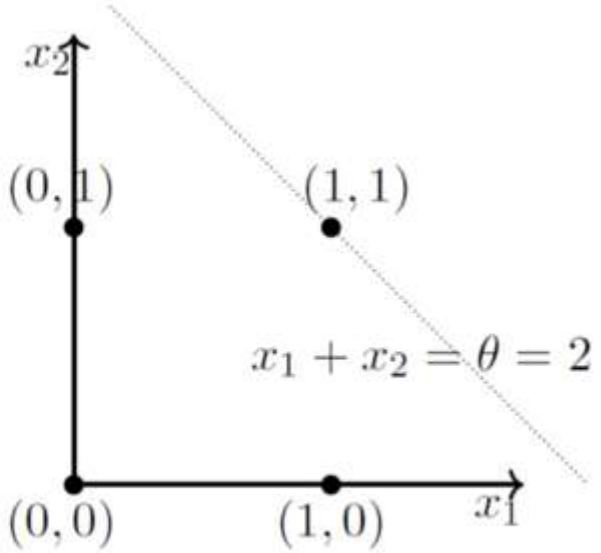
4. Geometric Interpretation Of M-P Neuron

AND Function



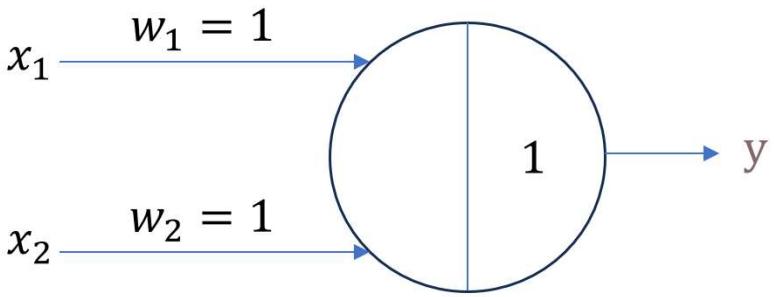
$$AND\ function$$

$$x_1 + x_2 = \sum_{i=1}^{2} x_i \geq 2$$



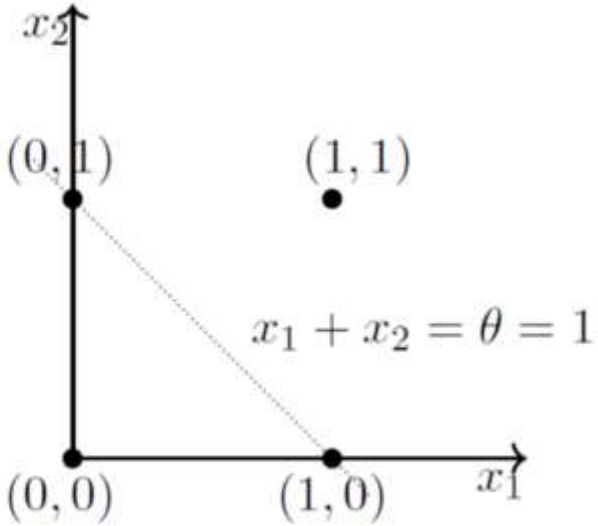
4. Geometric Interpretation Of M-P Neuron

OR Function



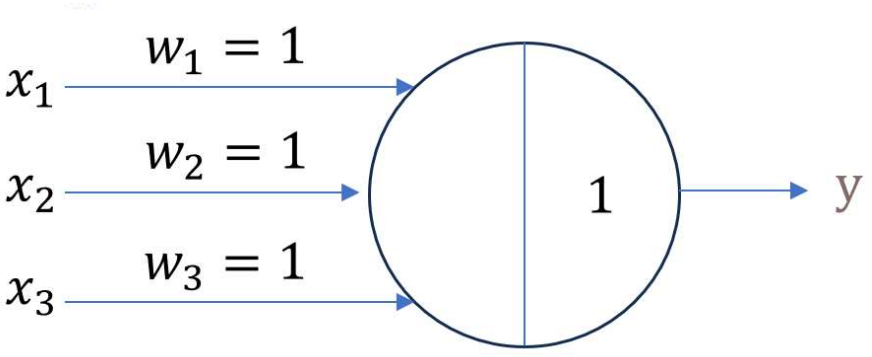
$$OR\ function$$

$$x_1 + x_2 = \sum_{i=1}^{2} x_i \geq 1$$



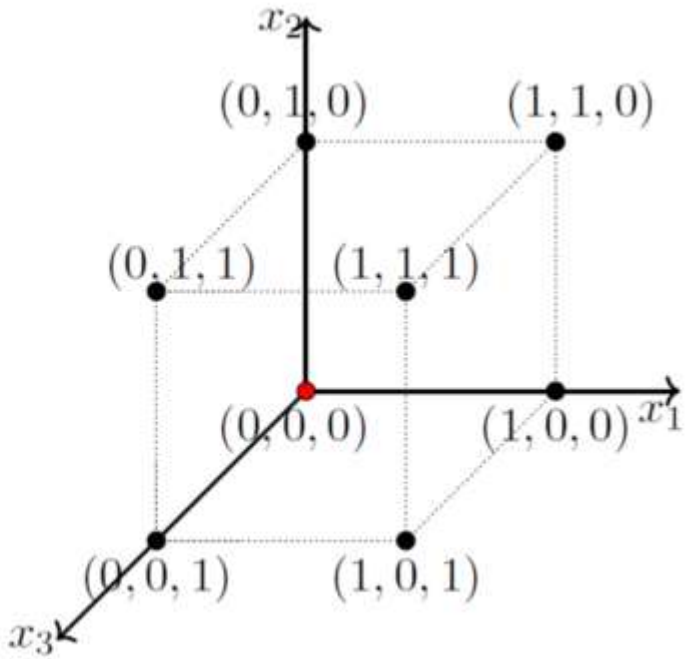
4. Geometric Interpretation Of M-P Neuron

OR Function With 3 Inputs

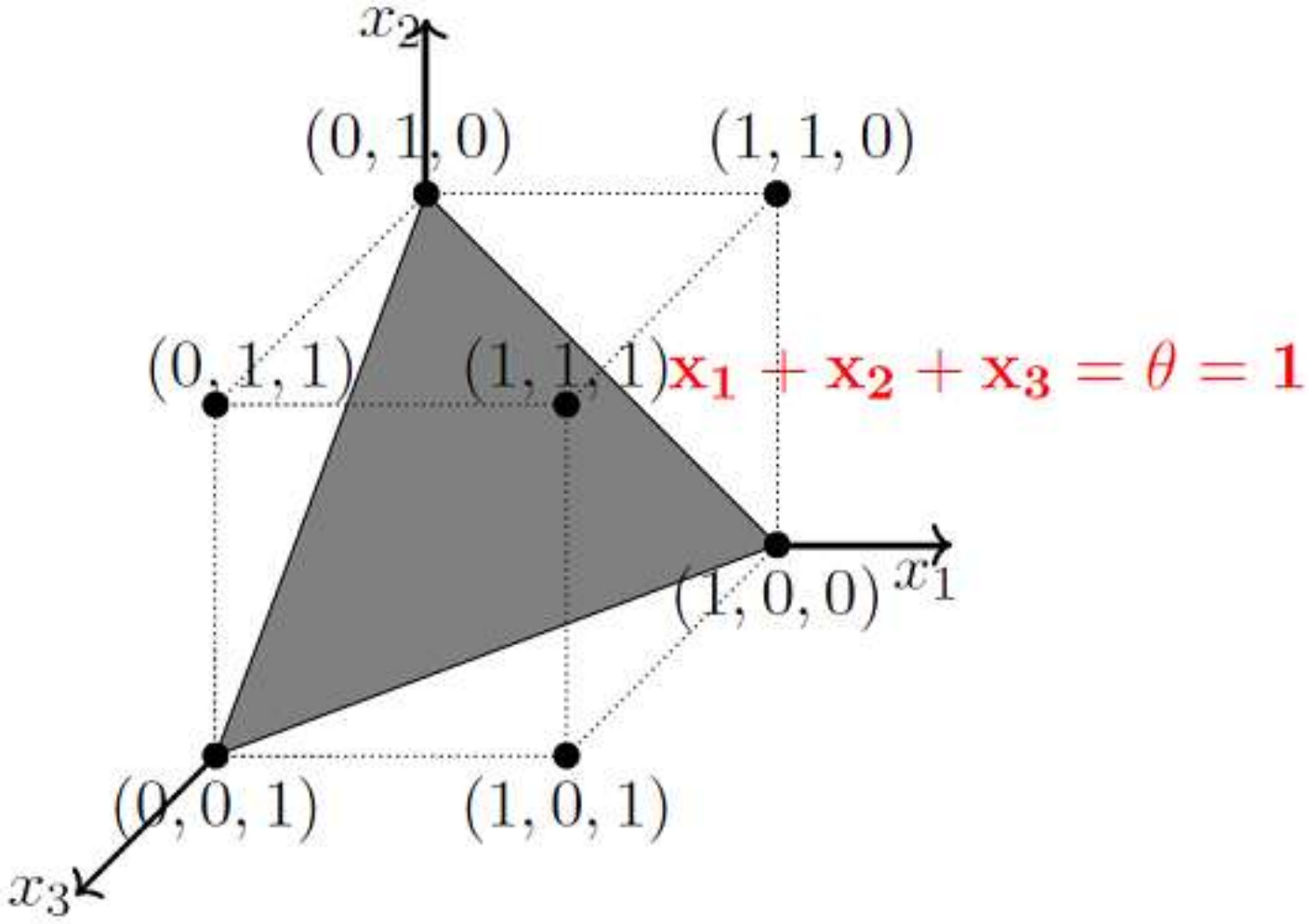


OR function

$$x_1 + x_2 + x_3 = \sum_{i=1}^{3} x_i \geq 1$$

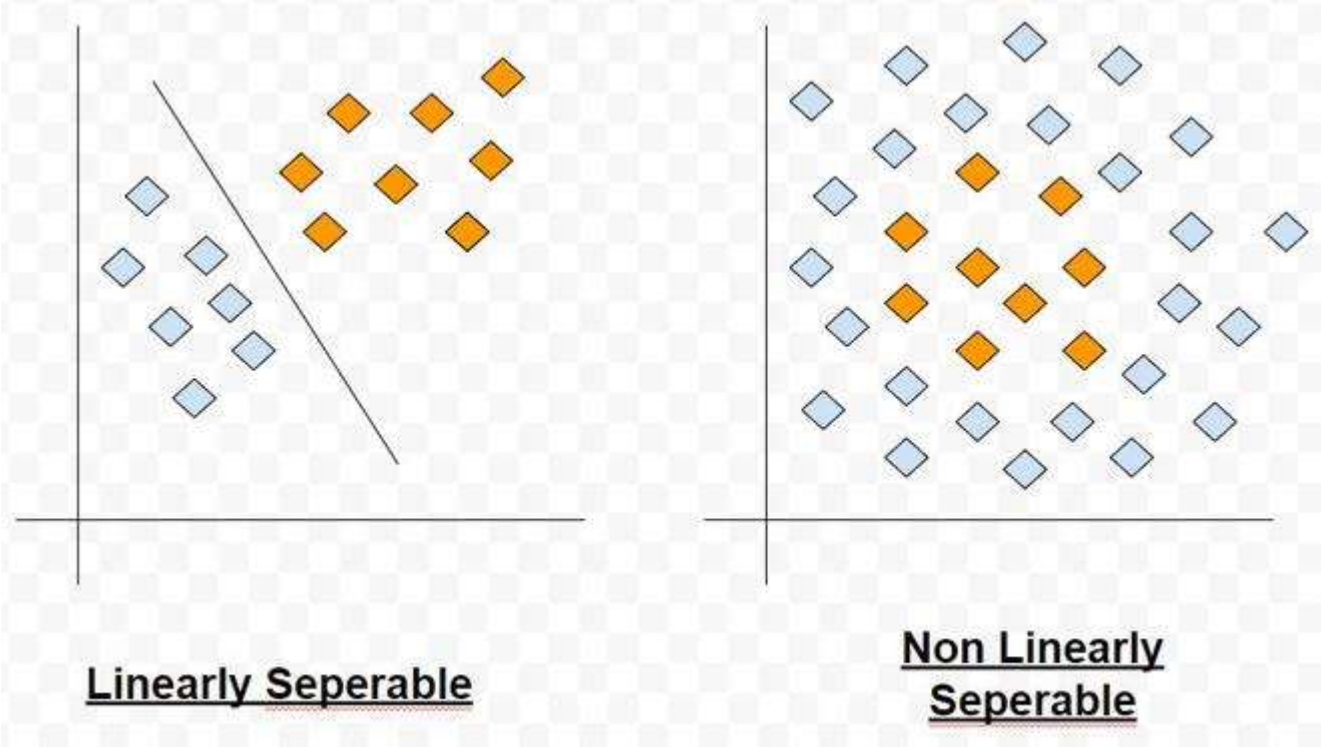


4. Geometric Interpretation Of M-P Neuron



4. Geometric Interpretation Of M-P Neuron

For single M-P Neuron, only linearly separable data can be classified





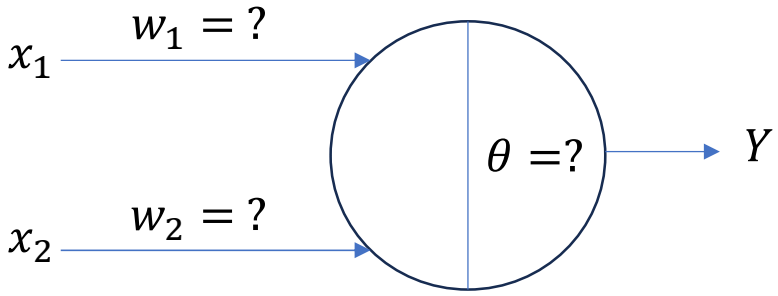
5. XOR

- Logic: The output is 1 if the number of 1s in the inputs is odd; otherwise, it is 0.

x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

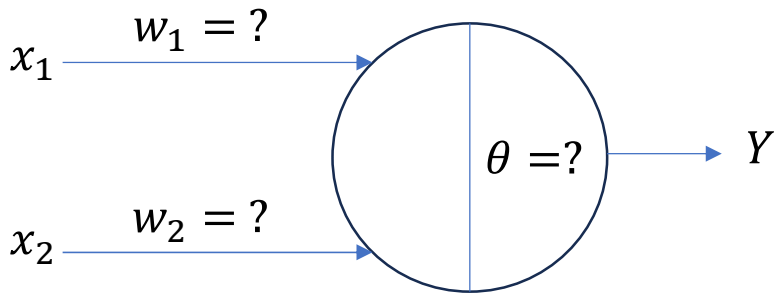
5. XOR

Can XOR be implemented using a single M-P Neuron? Explain Why?



x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

5. XOR



x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

w1*0 + w2*0 = $0 < \theta$ (1)

w1*0 + w2*1 = w2 $\geq \theta$ (2)

w1*1 + w2*0 = w1 $\geq \theta$ (3)

w1*1 + w2*1 = w1 + w2 $< \theta$ (4)

(2) + (3) $\Rightarrow$ $2\theta \leq$ w1 + w2

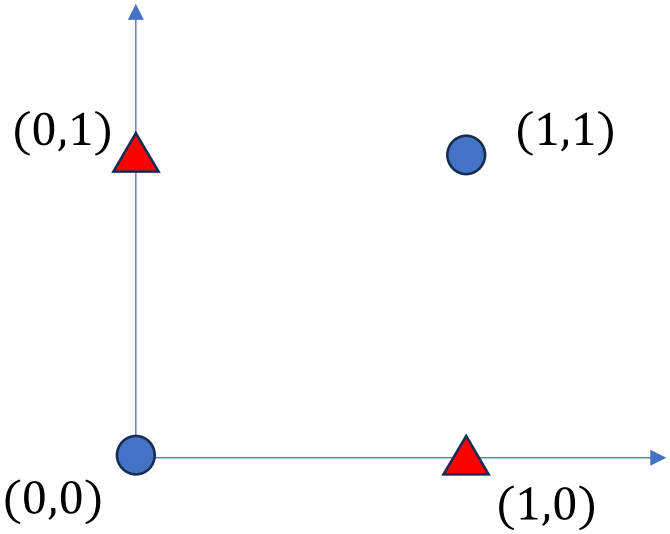
(4) $\Rightarrow$ w1 + w2 $< \theta$

$2\theta < \theta \Rightarrow \theta < 0$

5. XOR

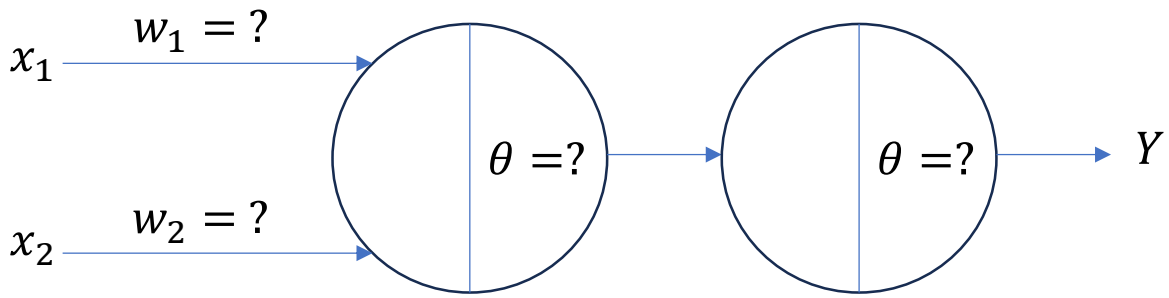
Geometric Interpretation

x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0



5. XOR

Can XOR be implemented using two M-P Neuron?

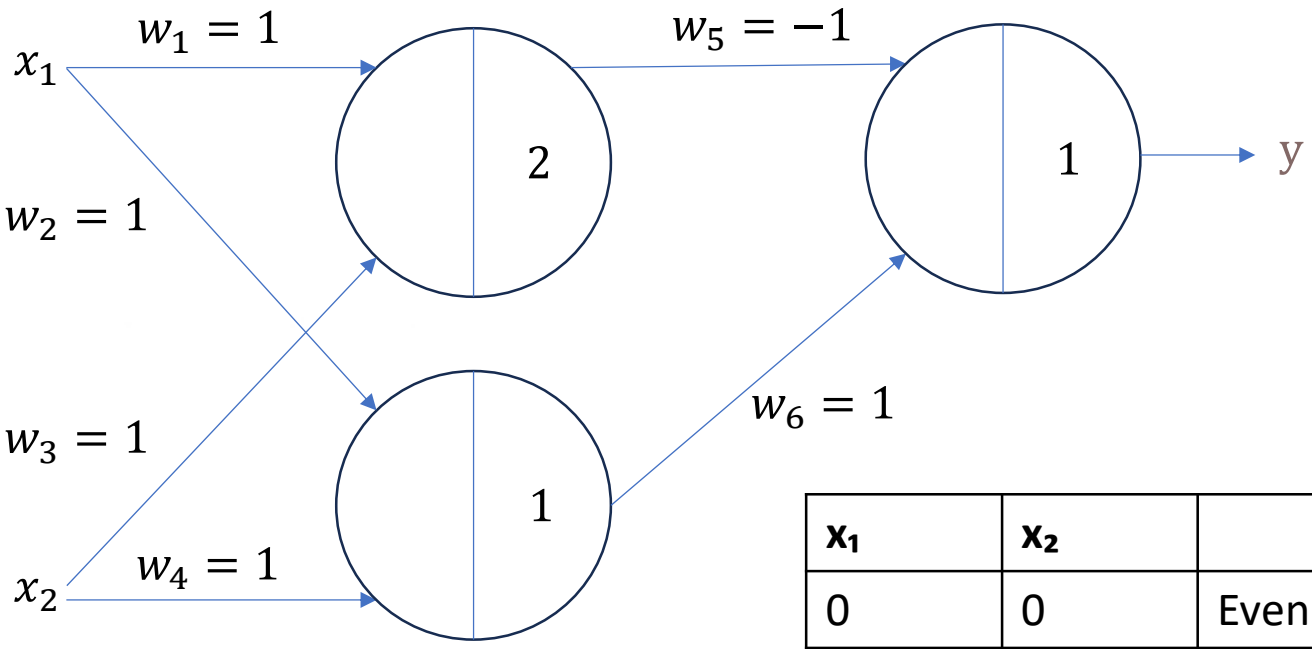


x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

5. XOR

Can XOR be implemented using three M-P Neuron?

- Logic: The output is 1 if the number of 1s in the inputs is odd; otherwise, it is 0.



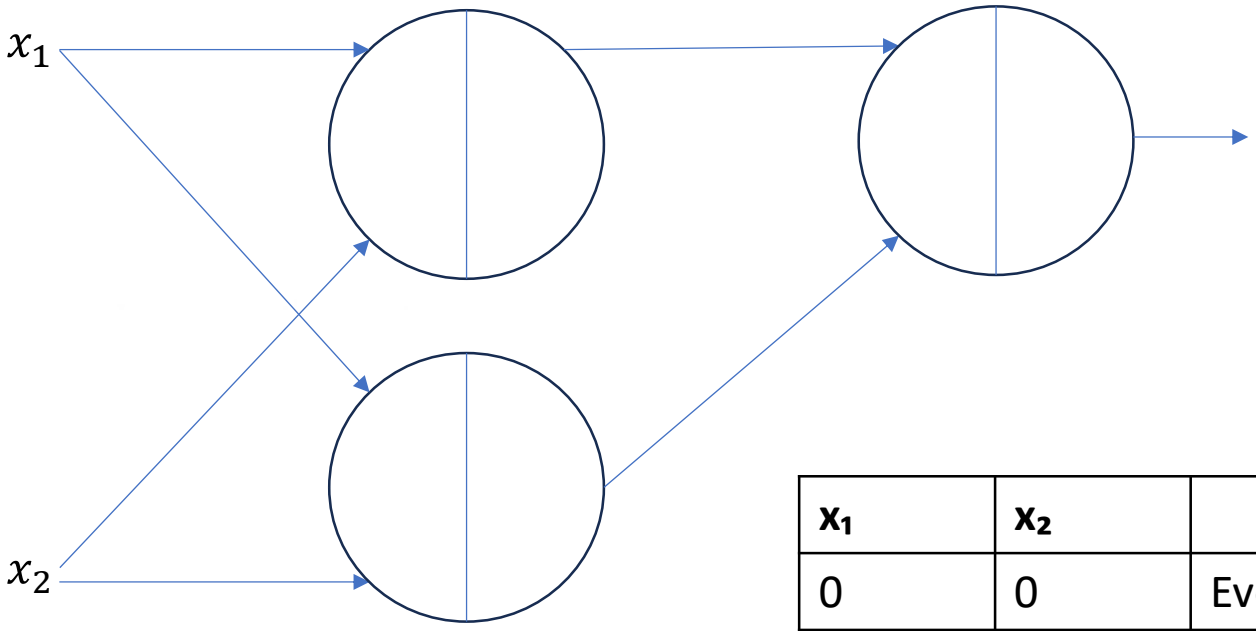
x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

Any other value for the threshold?

5. XOR

Any other solutions?

- Logic: The output is 1 if the number of 1s in the inputs is odd; otherwise, it is 0.



x_1	x_2		y (XOR)
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

5. XOR

3.7. XOR (3 inputs)

- Logic: The output is 1 if the number of 1s in the inputs is odd; otherwise, it is 0.

2 inputs

x_1	x_2		y
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

y	x_3		Z
0	0	Even	0
0	1	Odd	1
1	0	Odd	1
1	1	Even	0

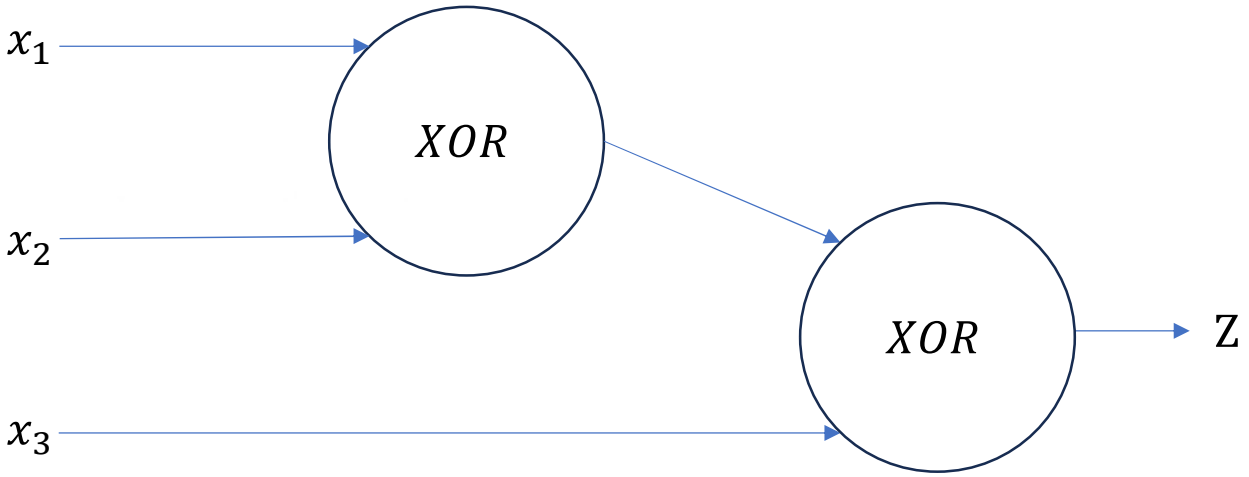
3 inputs

x_1	x_2	x_3		Z
0	0	0	Even	0
0	0	1	Odd	1
0	1	0	Odd	1
0	1	1	Even	0
1	0	0	Odd	1
1	0	1	Even	0
1	1	0	Even	0
1	1	1	Odd	1

5. XOR

3.6. XOR (3 inputs)

- Logic: The output is 1 if the number of 1s in the inputs is odd; otherwise, it is 0.

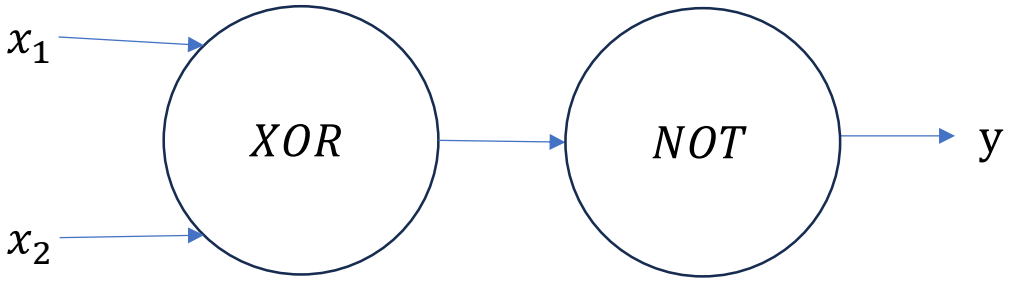


5. XOR

3.8. XNOR

- Logic: The output is 1 if the number of 1s in the inputs is even; otherwise, it is 0.
- Combine XOR gate with a NOR gate.

- Can you use only three M-P neurons to achieve XNOR?



x_1	x_2		y (XNOR)
0	0	Even	1
0	1	Odd	0
1	0	Odd	0
1	1	Even	1

Reference

McCulloch W S, Pitts W. A logical calculus of the ideas immanent in nervous activity[J]. The bulletin of mathematical biophysics, 1943, 5(4): 115-133.

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